

Linear Algebra (E)

2025 Fall

Final Exam

2026.1.11

Exam Duration: 3 Hours

Name: _____

Student ID: _____

This exam includes 7 pages (including this page) and 6 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam allows one SINGLE cheat sheet (size A4, double-sided is fine). No calculator or any electronic device is allowed. Plagiarism of all kinds is strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. At least write down one sentence to explain your thoughts.
- **Write clearly and legibly.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 62. No overflow.

Problem	Points	Score
1	6	
2	10	
3	11	
4	14	
5	12	
6	9	
Total:	62	

1. (Basic Problems)

These problems are basic.

- (a) (2 points) Suppose a 3×3 real matrix A has eigenvalues $\lambda_1 = -1$, $\lambda_2 = 1$, and $\lambda_3 = 2$.
- Find the determinant of A .
 - Find the trace of the matrix A^2 .

- (b) (2 points) Let B be a 3×3 matrix with $\det(B) = 4$. Calculate the value of $\det(2B^{-1})$.

- (c) (2 points) Find the value of k that makes the following three vectors in $\mathbb{R}^3$ linearly dependent:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ k \\ 1 \end{bmatrix}.$$

2. (The Variable Parameter)

Let C_x be a 4×4 matrix depending on a real scalar parameter x :

$$C_x = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 1 & x \\ 1 & 1 & -x & 1 \end{bmatrix}.$$

- (a) (2 points) Without expanding the full determinant, find the coefficient of the x^2 term in $\det(C_x)$.

- (b) (3 points) Using cofactor expansion or any method, show that the characteristic polynomial of C_x is:

$$p(\lambda) = (\lambda - 2)(\lambda - 1)(\lambda^2 - 2\lambda + x^2).$$

For what real values of x is the matrix C_x singular?

- (c) (3 points) Let $x = 2$. Find all 4 eigenvalues of C_2 . Find a complex eigenvector corresponding to the eigenvalue with a positive imaginary part.

- (d) (2 points) Let $M = (2I - C_x)(I - C_x)$ and $N = C_x^2 - 2C_x + x^2I$. Compute the matrix product MN . (Hint: Do NOT expand the matrices element-wise. Use the result from part (b)).

3. Consider the 5×5 matrix A , which has a symmetric pattern resembling an hour-glass:

$$A = \begin{bmatrix} a & 0 & 0 & 0 & b \\ 0 & a & 0 & b & 0 \\ 0 & 0 & a+b & 0 & 0 \\ 0 & b & 0 & a & 0 \\ b & 0 & 0 & 0 & a \end{bmatrix}.$$

- (a) (4 points) For generic a, b , find the spectral decomposition of A . That is, find an orthogonal matrix Q and a diagonal matrix Λ such that $A = Q\Lambda Q^T$.
- (b) (2 points) For what conditions of a, b will this matrix be indefinite?
- (c) (3 points) If $a = b \neq 0$, find a basis for the vector space $V = \{p(A) : p \text{ is any polynomial}\}$. (Remember to prove spanning and linear independence for your basis elements.)
- (d) (2 points) For $a = b = 1$, find a polynomial $p(x)$ of degree 1 such that $(A + I)^{-1} = p(A)$.

4. We are trying to recover a signal $\mathbf{x} \in \mathbb{R}^2$ from a measurement $\mathbf{b} \in \mathbb{R}^3$. The measurement process is modeled by $A\mathbf{x} = \mathbf{b}$, where:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ t \end{bmatrix}.$$

However, the vector $\mathbf{b}$ has been corrupted by noise and is not in the range of A .

- (a) (3 points) Find the singular values of A .
- (b) (4 points) When is $\mathbf{b}$ not in $\text{Ran}(A)$? When $\mathbf{b} \notin \text{Ran}(A)$, find the QR decomposition of the 3×3 matrix $\begin{bmatrix} A & \mathbf{b} \end{bmatrix}$ when $t > 4$. (I picked $t > 4$ for no particular reason. It will streamline the process to make the diagonal entries of R positive.)
- (c) (2 points) Find the orthogonal projection matrix P onto the subspace $\text{Ran}(A)$.
- (d) (2 points) Find the least squares solution $\hat{\mathbf{x}}$ that minimizes $\|A\mathbf{x} - \mathbf{b}\|^2$.
- (e) (3 points) (HARD) Let S be the unit circle in $\mathbb{R}^2$. Let $B = \begin{bmatrix} 2 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. The image $B(S)$ is an ellipse in $\mathbb{R}^3$. Find the lengths of the long and short half-axes of this ellipse, and find the vectors in $\mathbb{R}^3$ that point along these axes.

5. Let A be a 2×2 real matrix. We construct a larger 4×4 block matrix K using polynomials of A :

$$K = \begin{bmatrix} A & A + I \\ A + I & A + 2I \end{bmatrix}.$$

- (a) (2 points) Calculate the determinant of K . (Hint: Use block Gaussian elimination to simplify the matrix structure).

- (b) (3 points) Find the inverse matrix K^{-1} . Express your answer in block form in terms of A and I .

- (c) (3 points) Let $\mathbf{v}$ be an eigenvector of A with eigenvalue λ (assume $\lambda \neq -1$). Find all scalars α such that the vector $\mathbf{w} = \begin{bmatrix} \mathbf{v} \\ \alpha \mathbf{v} \end{bmatrix}$ is an eigenvector of K .

- (d) (2 points) Now let $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Find all 4 eigenvalues of K .

- (e) (2 points) Under what condition of A , will the matrix K be positive definite?

6. Let V be the vector space of all 2×2 real matrices with trace equal to zero. We consider the linear transformation $L : V \rightarrow V$ defined by the commutator with a nilpotent matrix:

$$L(X) = JX - XJ, \quad \text{where } J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Let $\mathcal{B} = (E, H, F)$ be the ordered basis for V , where:

$$E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad F = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

- (a) (3 points) Find the matrix representation of L under the basis $\mathcal{B}$.
- (b) (3 points) Find all eigenvalues of L along with their algebraic and geometric multiplicities. Is L diagonalizable?
- (c) (3 points) Consider a new ordered basis $\mathcal{C} = (E', H', F')$ where $E' = E$, $H' = E + H$, and $F' = E + H + F$. Find the change of coordinate matrix $P_{\mathcal{B} \leftarrow \mathcal{C}}$ from $\mathcal{C}$ to $\mathcal{B}$. Then, using this matrix, calculate the matrix representation of L under the new basis $\mathcal{C}$.